MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. If $\tan\theta = 1$, then the value of $\frac{8\sin\theta+5\cos\theta}{\sin^3\theta-2\cos^3\theta+7\cos\theta}$ is: [1]

(A) 2 (B) $2\frac{1}{2}$ (C) 3 (D) $\frac{4}{5}$

Q2. The maximum value of $\frac{1}{\csc\theta}$ for $0° < \theta \leq 90°$ is: [1]

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

Q3. The value of $(1+\tan^2\theta)(1-\sin\theta)(1+\sin\theta)$ is: [1]

(A) 0 (B) 1 (C) -1 (D) $\sec^2\theta$

Q4. The value of $9\sec^2 A - 9\tan^2 A$ is equal to: [1]

(A) 1 (B) 9 (C) 8 (D) 0

Q5. $(1+\tan\theta+\sec\theta)(1+\cot\theta-\csc\theta)$ is equal to: [1]

(A) 0 (B) 1 (C) 2 (D) -1

Q6. **Assertion (A):** The value of $\tan 60° \cot 60° = 1$.
Reason (R): $\tan\theta$ and $\cot\theta$ are reciprocals of each other for any valid acute angle θ. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If $15\cot A = 8$, find the values of $\sin A$ and $\sec A$. [2]

Q8. Evaluate: $2\tan^2 45° + \cos^2 30° - \sin^2 60°$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. If $a\cos\theta - b\sin\theta = c$, prove that $a\sin\theta + b\cos\theta = \pm\sqrt{a^2 + b^2 - c^2}$. [3]

Q10. If $\csc\theta + \cot\theta = q$, prove that $\cos\theta = \frac{q^2-1}{q^2+1}$. [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

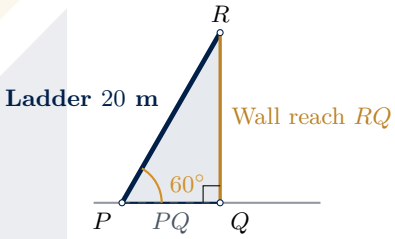
Q11. Prove that: $\frac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = \frac{1}{\sec\theta - \tan\theta}$, using the identity $\sec^2\theta = 1 + \tan^2\theta$. [5]

Q12. If $a\cos\theta + b\sin\theta = m$ and $a\sin\theta - b\cos\theta = n$, prove that $a^2 + b^2 = m^2 + n^2$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Fire Brigade Aerial Turntable Ladder Rescue Operation

A municipal fire rescue truck carries an aerial turntable ladder. During a rescue drill, the ladder base is mounted on the truck platform at point P. The ladder PR of maximum extension length 20 m is inclined at an angle of $\theta = 60°$ to the horizontal to reach a window ledge R of a burning building.



Based on the above geometric model, answer the following questions:

(i) Calculate the vertical reach height RQ of the ladder above the truck platform level. [1]

(ii) Find the horizontal distance PQ from the ladder turntable to the building wall. [1]

(iii) If safety regulations permit a maximum inclination angle of 45° in high winds, what would be the vertical reach height of the ladder at 45° extension of 20 m? [2]

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